

Disproof of TLMC Conjecture 00000000521: Alternating Betti Sums of Edge Ideals

TLMC falsification submission

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Abstract

We disprove TLMC conjecture 00000000521, which claims that for every edge ideal I the alternating Betti sums $A_j = \sum_i (-1)^i \beta_{i,j}(I)$ change sign exactly once as j crosses $\text{reg}(I)$, with $|A_j|$ monotonically nondecreasing up to that point. The triangle graph C_3 with $I = (xy, xz, yz) \subset k[x, y, z]$ is a counterexample: its alternating sums are $A_0 = 1$, $A_1 = 0$, $A_2 = -3$, $A_3 = 2$, whose signs $+, -, +$ change twice. Even a path of two adjacent edges fails, and the conjecture fails under either convention for reading $\beta_{i,j}(I)$. All numbers are independently recomputed (Taylor resolution tensored with k , cross-checked against Hilbert series) and formalized in axiom-free Lean 4.

1 The conjecture

For a graded module let $\beta_{i,j}$ denote its graded Betti numbers and put

$$A_j = \sum_{i \geq 0} (-1)^i \beta_{i,j}.$$

Conjecture 00000000521 asserts that for every edge ideal I of a simple graph the sequence $\{A_j\}$ changes sign *exactly once* as j crosses $\text{reg}(I)$, and that $|A_j|$ is monotonically nondecreasing up to that point; this was proposed as a combinatorial characterization of the graded Lefschetz property for edge ideals.

2 The counterexample

Let $G = C_3$ be the triangle and let $I = I(G) = (xy, xz, yz) \subset k[x, y, z]$. The minimal free resolution of R/I is

$$0 \longrightarrow R(-3)^2 \longrightarrow R(-2)^3 \longrightarrow R \longrightarrow R/I \longrightarrow 0,$$

with syzygies $(z, -y, 0)$, $(z, 0, -x)$, $(0, y, -x)$ on the three generators and a single syzygy among them, so the syzygy module is free of rank 2. Exactness is witnessed by the Hilbert series:

$$\frac{1 - 3t^2 + 2t^3}{(1 - t)^3} = \frac{(1 - t)^2(1 + 2t)}{(1 - t)^3} = \frac{1 + 2t}{1 - t} = 1 + \frac{3t}{1 - t},$$

matching $R/I \simeq k \oplus \bigoplus_{d \geq 1} (kx^d \oplus ky^d \oplus kz^d)$. Hence the Betti table of I (Betti numbers of R/I) is

	$j = 0$	$j = 2$	$j = 3$
$i = 0$	1		
$i = 1$		3	
$i = 2$			2

and therefore

$$A_0 = 1, \quad A_1 = 0, \quad A_2 = -3, \quad A_3 = 2, \quad A_j = 0 \ (j \geq 4).$$

Also $\text{reg}(I) = 2$: the resolution of I itself is $0 \rightarrow R(-3)^2 \rightarrow R(-2)^3 \rightarrow I \rightarrow 0$, so $\beta_{0,2}(I) = 3$, $\beta_{1,3}(I) = 2$ and $\text{reg}(I) = \max(j - i) = 2$.

Theorem 1 (Falsification). *Conjecture 00000000521 is false. For $I = (xy, xz, yz)$ the nonzero entries of $\{A_j\}$ are $1, -3, 2$, i.e. the signs run $+, -, +$: the sign changes twice, not exactly once, and $|A_0|, |A_2|, |A_3| = 1, 3, 2$ is not monotonically nondecreasing.*

Remark 2 (Index correction). *The original falsification-queue entry recorded the attack numbers as $A_2 = 3, A_4 = -3, A_6 = 1$, which are index-shifted. The correct numbers, recomputed independently here (and in agreement with the verdict line), are $A_0 = 1, A_2 = -3, A_3 = 2$; the conclusion of the falsification is unchanged.*

3 Boundary: how widespread the failure is

All computations below come from the accompanying `reproduce.py` (see Section 4); every example passed the Hilbert-series cross-check.

- The failure is minimal: already $I = (xy, yz)$ (a path of two adjacent edges in $k[x, y, z]$, resolution $0 \rightarrow R(-3) \rightarrow R(-2)^2 \rightarrow R \rightarrow R/I \rightarrow 0$) gives $A = (1, 0, -2, 1)$: two sign changes.
- Disjoint edges: $I = (xy, zt) \subset k[x, y, z, t]$ gives $A = (1, 0, -2, 0, 1)$: two sign changes. P_4 gives $(1, 0, -3, 2, 0)$; the star $K_{1,3}$ gives $(1, 0, -3, 3, -1)$ and C_4 gives $(1, 0, -4, 4, -1)$ (three changes).
- Exhaustive scan: among all 63 labeled graphs on 4 vertices with at least one edge, **57** violate the “exactly one sign change” condition.
- The failure is robust to the convention for $\beta_{i,j}(I)$. If instead one resolves I as a module, then $A_j(I) = \beta_{0,j}(R/I) - A_j(R/I)$, and: C_3 gives $A(I) = (0, 0, 3, -2)$ (one change, no violation), but C_4 gives $A(I) = (0, 0, 4, -4, 1)$ and $K_{1,3}$ gives $(0, 0, 3, -3, 1)$: two sign changes each; a single edge gives $A(I) = (0, 0, 1)$, which never changes sign. No convention rescues the conjecture.

4 Verification

Independent recomputation (`reproduce.py`). The script builds the Taylor resolution of R/I , tensors it with $k = \text{GF}(10^9 + 7)$, and computes the homology dimensions per bidegree, which yields $\beta_{i,j}(R/I)$ (Tor is computable from any free resolution; after tensoring with k only the constant entries of the Taylor differentials survive). It then cross-checks A_j against the coefficient of t^j in $(1 - t)^n \cdot \text{Hilb}_{R/I}(t)$ obtained by direct monomial enumeration, for every example and every graph in the scan. Run with `python3 reproduce.py`.

Lean 4 formalization (lean4/). The Betti table $\beta_{0,0} = 1$, $\beta_{1,2} = 3$, $\beta_{2,3} = 2$, the alternating sums $A_0 = 1$, $A_1 = 0$, $A_2 = -3$, $A_3 = 2$, the *two* sign changes, and the failure of monotonicity of $|A_j|$ are proved by `decide` in pure Lean 4 core (v4.33.1, no Mathlib). `lake build && lake env lean Check.lean` succeeds, and every `#print axioms` reports “*does not depend on any axioms*”; there are no `sorry` and no additional axioms.